10.4 Mann-Whitney-Wilcoxon Procedure

① Mann-Whitney-Wilcoxon statistic

(i) problem

$X_1, \cdots, X_{n_1}$: r.s. from a distribution with cdf $F(x)$

$Y_1, \cdots, Y_{n_2}$: r.s. from a distribution with cdf $F(x-\Delta)$

We are interested in testing $H_0 : \Delta = 0 \quad H_1 : \Delta > 0$

(ii) MWW statistic

Reject $H_0 : \Delta = 0$ if $W = \sum_{j=1}^{n_2} R(y_j) \geq c$, where $R(Y_j)$ is the rank of Y_j in the conbined sample of size $n = n_1 + n_2$.

(iii) Theorem 10.4.1.

(a) W is distribution free with a symmetric pdf

(b) $E_{H_0}(W) = n_2(n+1)/2$

(c) $Var_{H_0}(W) = n_1 n_2 (n+1)/12$

(d) $\dfrac{W - \frac{n_2(n+1)}{2}}{\sqrt{n_1 n_2 (n+1)/12}} \xrightarrow{D} N(0,1)$

(pf)

(a) Since $P(R(Y_1) = r_1, \cdots, R(Y_{n_2}) = r_{n_2}) = 1/\binom{n}{n_2}$, W is distribution free.

(b) Note that $R(Y_j)$ is uniformly distributed on $\{1, \cdots, n\}$ under H_0, we have

$$E(W) = \sum_{j=1}^{n_2} E(R(Y_j)) = \sum_{j=1}^{n_2} \left\{ \sum_{i=1}^{n} i \cdot \frac{1}{n} \right\} = n_2 \cdot \frac{1}{n} \cdot \frac{n(n+1)}{2} = \frac{n_2(n+1)}{2}$$

(c) is quite tedious to derive, and (d) is beyond our scope.

(iv) another formulation of W.

First, note that $R(Y_j) = \sum_{i=1}^{n_1} I(X_i < Y_j) + \sum_{k=1}^{n_2} I(Y_k < Y_j)$, therefore,

$$W = \sum_{j=1}^{n_2} R(Y_j) = \sum_{i=1}^{n_1}\sum_{j=1}^{n_2} I(X_i < Y_j) + \sum_{k=1}^{n_2}\sum_{j=1}^{n_2} I(Y_k < Y_j) \equiv U + \frac{n_2(n_2+1)}{2}$$

where $U = \sum_i \sum_j I(X_i < Y_j)$.

Hence, in testing $\Delta = 0$, reject $H_0 : \Delta = 0$ if $U \geq c$. Also, note that

$$E(U) = E(W) - \frac{n_2(n_2+1)}{2} = \frac{n_2(n+1)}{2} - \frac{n_2(n_2+1)}{2} = n_1 n_2 / 2.$$

To compute power under $H_1 : \Delta > 0$, we have

$$U(\Delta) = \sum_i \sum_j I(Y_j - \Delta > X_i) = \sum_i \sum_j I(Y_j - X_i > \Delta)$$

② ARE

(i) efficacy of U.

Consider $H_0 : \Delta = 0, \quad H_{1n} : \Delta_n = \delta/\sqrt{n}, \ \ \delta > 0$, and assume that $n_1/n \to \lambda_1$, $n_2/n \to \lambda_2$ s.t. $\lambda_1 + \lambda_2 = 1$.

To compute efficacy of U, need to compute $\mu_U(\Delta_n)$ and $Var_{H_0}(U)$.

Now,

$$\mu_U(\Delta_n) = E_{\Delta_n}(U(0)) = E_0(U(-\Delta_n)) = n_1 n_2 P(Y - X > -\Delta_n)$$

$$\begin{aligned} P(Y - X > -\Delta_n) &= E_0[P(Y - X > -\Delta_n | X)] = E_0[P(Y > X - \Delta_n | X)] \\ &= E_0[1 - F(X - \Delta_n)] = 1 - \int F(x - \Delta_n) f(x) dx \\ &= \frac{1}{2} + \int [F(x) - F(x - \Delta_n)] f(x) dx \simeq \frac{1}{2} + \Delta_n \int f^2(x) dx \end{aligned}$$

$$\therefore \mu_U(\Delta_n) = n_1 n_2 \{\tfrac{1}{2} + \Delta_n \textstyle\int f^2(x)dx\} \to \mu'_U(0) = n_1 n_2 \textstyle\int f^2(x)dx$$

$$Var_{H_0}(U) = n_1 n_2 (n+1)/12$$

Hence, the efficacy of U is

$$c_U = \lim_{n\to\infty} \frac{\mu'_U(0)}{\{nVar(U)\}^{1/2}} = \lim_{n\to\infty} \frac{n_1 n_2 \int f^2(x)dx}{\sqrt{n_1 n_2 n(n+1)/12}} = \sqrt{12}\sqrt{\lambda_1\lambda_2} \int f^2(x)dx$$

(ii) ARE(U, t) at normal distribution

$X_1, \cdots, X_{n_1}$: r.s. from $N(0,1)$

$Y_1, \cdots, Y_{n_2}$: r.s. from $N(\Delta, 1)$

X, Y : indep, $\Delta > 0$.

$$H_0 : \Delta = 0 \qquad H_1 : \Delta > 0$$

The t-test stat. is $t = \dfrac{\bar{Y} - \bar{X}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ i.e. reject H_0 if $\bar{Y} - \bar{X} \geq c$.

$\mu_t(\Delta_n) = E_{\Delta_n}(\bar{Y} - \bar{X}) = \Delta_n \ \to \ \mu'_t(0) = 1.$

$Var(t) = Var_{H_0}(\bar{Y} - \bar{X}) = \frac{1}{n_1} + \frac{1}{n_2}$.

$\therefore$ the efficacy of t is $c_t = \lim_{n\to\infty} \frac{\mu'_t(0)}{\{nVar(t)\}^{1/2}} = \lim_{n\to\infty} \frac{1}{\sqrt{n\cdot(\frac{1}{n_1}+\frac{1}{n_2})}} =$

$\lim_{n\to\infty} \frac{1}{\sqrt{n\frac{n_1+n_2}{n_1 n_2}}} = \lim_{n\to\infty} \sqrt{\frac{n_1}{n}\frac{n_2}{n}} = \sqrt{\lambda_1\lambda_2}$

Also, we derived efficacy of U as $c_U = \sqrt{12}\sqrt{\lambda_1\lambda_2} \int f^2(x)dx$

$\therefore$ ARE(U,t)$= \dfrac{c_U^2}{c_t^2} = 12\left(\int f^2(x)dx\right)^2 = \dfrac{3}{\pi} \simeq .955$

i.e. MWW is 95.5% as efficient as t-test